Encoding Categorical: One Hot Encoding

Hazem Mohamed
AI Engineer

Encoding Categorical

1. DEFINITION

Hazem Mohamed
AI Engineer

Encoding Categorical

- Encoding Categorical: is used when we have categorical (non numerical data), such as countries: Germany, France, Span
- Note:

Machine learning models cannot work directly with text, sp we convert these categories into numerical form

Hazem Mohamed
AI Engineer

One Hot Encoding

EXPLAIN WITH EXAMPLE

Hazem Mohamed
AI Engineer

One Hot Encoding

- One Hot Encoding:
  - Instead of assigning numbers like :
    - Germany = 1
    - France = 2
    - Spain = 3
- ❌ This is wrong because it creates a false order
  - (Spain > France > Germany)
- ✅ Correct Approach: One Hot Encoding
  - We create a separate column for each category:

Hazem Mohamed
AI Engineer

One Hot Encoding

Country	Germany	France	Spain
Germany	1	0	0
France	0	1	0
Spain	0	0	1

- Each row has:
  - 1 → The country exists
  - 0 → The country does not exist
- These columns are called dummy variables.
- Why are they called "Dummy Variables?
  - In statistics and machine learning, the term "dummy" doesn't mean "stupid." Instead, it refers to something that is a stand-in or a placeholder for something else.

Hazem Mohamed
AI Engineer

One Hot Encoding

- Final clean Definition:
  - One Hot Encoding is a technique used to convert categorical variable into multiple binary (0 / 1) columns, where each category is represented by a separate columns, allowing machine learning model to process non numeric data without introducing false relationship

Hazem Mohamed
AI Engineer

Follow for More

AI | Machine Learning | Python

- Simple explanations + Practical examples

Follow for more

Hazem Mohamed
AI Engineer